

RL

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1 Introduction

Action-value:

$$\begin{aligned} Q(s, a) &= \mathbb{E}[G_t | S_t = s, A_t = a] \\ &= \mathbb{E}[R_{t+1} + \gamma G_{t+1} | S_t = s, A_t = a] \\ &= \mathbb{E}[R_{t+1} | S_t = s, A_t = a] + \gamma \mathbb{E}[G_{t+1} | S_t = s, A_t = a] \\ &= \mathbb{E}[R_{t+1} | S_t = s, A_t = a] + \gamma \sum_{s' \in \mathcal{S}} \mathbb{E}[G_{t+1} | S_t = s, A_t = a, S_{t+1} = s'] P(S_{t+1} = s' | S_t = s, A_t = a) \\ &= \mathbb{E}[R_{t+1} | S_t = s, A_t = a] + \gamma \sum_{s' \in \mathcal{S}} V(s') P(S_{t+1} = s' | S_t = s, A_t = a) \\ &= \mathbb{E}[R_{t+1} | S_t = s, A_t = a] + \gamma \mathbb{E}[V(S_{t+1}) | S_t = s, A_t = a] \\ &= \mathbb{E}[R_{t+1} + \gamma V(S_{t+1}) | S_t = s, A_t = a] \end{aligned}$$

State-value in terms of action-value:

$$\begin{aligned} V_\pi(s) &= \mathbb{E}[G_t | S_t = s] \\ &= \sum_{a \in \mathcal{A}} \mathbb{E}[G_t | S_t = s, A_t = a] P(A_t = a | S_t = s) \\ &= \mathbb{E}_{a \sim \pi} [Q_\pi(s, a)] \end{aligned}$$

Action-value in terms of state-value (from above):

$$\begin{aligned} Q_\pi(s, a) &= \mathbb{E}[R_{t+1} | S_t = s, A_t = a] + \gamma \sum_{s' \in \mathcal{S}} V(s') P_{ss'}^a \\ &= R(s, a) + \gamma \sum_{s' \in \mathcal{S}} V(s') P_{ss'}^a \end{aligned}$$

Bellman Optimality Equations. State-value:

$$\begin{aligned} V_*(s) &= \max_{\pi} V_\pi(s) = \max_{\pi} \sum_{a \in \mathcal{A}} Q_\pi(s, a) \pi(a | s) \\ &= \max_{a \in \mathcal{A}} Q_*(s, a) \end{aligned}$$

Action-value:

$$\begin{aligned} Q_*(s, a) &= R(s, a) + \gamma \max_{\pi} \sum_{s' \in S} V_{\pi}(s') P_{ss'}^a \\ &= R(s, a) + \gamma \sum_{s' \in S} V_*(s') P_{ss'}^a \end{aligned}$$